M M Abid Naziri

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About Me

PhD Researcher specializing in **Software Testing** for learning-enabled systems. I develop automated testing techniques for Deep Learning Libraries (**PyTorch**, **TensorFlow**), **Autonomous Driving Systems** in a simulation environment (**CARLA**). I have reported **323** bugs (**214** confirmed) across 11 software systems to date [list].

Education

NC State University, PhD in Computer Science (5th year) May 2027 (Expected)

Advisor: Dr. Marcelo d'Amorim

Research Projects & Publications

A Multi-Domain Study on the Effectiveness of LLMs at Writing Property-Based Tests *Submitted*

M M Abid Naziri, Feiran Qin, Khush Patel, Marcelo d'Amorim

- Studied how well LLMs generalize as Property-Based Test (PBT) writers across **10** applications from **5** domains: Deep Learning libraries, databases, system software, compilers, and LLM serving engines.
- Filed **180** bug reports from the generated PBTs; maintainers confirmed **65** across all **5** domains (**23** Fixed).
- Found that low observability is the main obstacle for LLM-based PBT writers, and that domain-specific fuzzers still cover more of the compilers and of PostgreSQL.

Preprint: arXiv [pdf]

A Comprehensive Study of Bugs from Deep Learning Library Fuzzers *Submitted*

M M Abid Naziri, Manideepika Reddy Myaka, Lingjun Liu, Saikat Dutta, Marcelo d'Amorim

- Studied **220** bugs from **5** state-of-the-art Deep Learning Library testing techniques; found **70%** are validation bugs from simple invalid-input patterns, not deep logic errors.
- Built a mutational fuzzing tool from these patterns, uncovering **46** new bugs in these libraries (**30** confirmed).

How Effective Is Coverage-Guided Fuzzing to Test Deep Learning Library APIs? ICST 2026

Feiran Qin, ***M M Abid Naziri***, Hengyu Ai, Saikat Dutta, Marcelo d'Amorim

- Conducted a study to assess the effectiveness of coverage guided fuzzing to test Deep Learning Library APIs.
- Reported **42** new bugs in Tensorflow and PyTorch (**8** Fixed).

Venue: IEEE International Conference on Software Testing, Verification and Validation (ICST 2026) [pdf]
Award: IEEE Computer Society TCSE Best Student Paper Award

Testing Deep Learning Libraries via Neurosymbolic Constraint Learning ICSE 2026

M M Abid Naziri[*], Shinhae Kim[*], Feiran Qin, Saikat Dutta, Marcelo d'Amorim

[*]Equal contribution

- Designed and implemented **Centaur**, a neurosymbolic constraint learning tool for DL APIs to generate valid test inputs.
- Uncovered **23** new bugs in Tensorflow and PyTorch (**11** confirmed).

Venue: IEEE/ACM International Conference on Software Engineering (ICSE 2026) [pdf]

Misbehavior Forecasting for Focused Autonomous Driving Systems Testing ICSE 2026

M M Abid Naziri, Stefano Carlo Lambertenghi, Andrea Stocco, Marcelo d'Amorim

- Developed **Foresee**, a tool to enhance simulation-based testing of self-driving software by identifying and analyzing near-miss scenarios, improving failure detection efficiency and effectiveness.
- **Foresee** exposes **128%** more failures than state-of-the-art failure predictors.
- **Foresee** also enhances the capability of an existing technique DriveFuzz by upto **94%**.

Venue: IEEE/ACM International Conference on Software Engineering (ICSE 2026) [pdf]

Improving Deep Learning Library Testing with Machine Learning AST 2026

Facundo Molina, ***M M Abid Naziri***, Feiran (Alex) Qin, Alessandra Gorla, Marcelo d'Amorim

- Applied ML classifiers to infer the validity of an input before executing to increase efficiency of DL Library fuzzing tools.
- The classifiers achieved an accuracy of **91**% in predicting input validity.
- They also improved an existing fuzzing tool called ACETest by increasing its validity ratio from **29**% to **61**%.

Venue: ACM/IEEE International Conference on Automation of Software Test (AST 2026) [pdf]

BugsInDLLs: A Database of Reproducible Bugs in Deep Learning Libraries to Enable Systematic Evaluation of Testing Techniques ISSTA 2025

M M Abid Naziri, Aman Kumar Singh, Benjamin Wu, Feiran (Alex) Qin, Saikat Dutta, Marcelo d'Amorim

- Developed an open-source dataset **BugsInDLLs** with **112** reproducible bugs across three popular DL libraries: JAX, Tensorflow, and PyTorch.
- Added support for benchmarking other fuzzing tools with an included integration of FreeFuzz.

Venue: International Symposium on Software Testing and Analysis (ISSTA 2025: Tool Demos) [pdf]

Experience

Senior Software Engineer, Enosis Solutions – Dhaka, Bangladesh Mar 2019 – Jul 2022

- Developed scalable web applications using Vue JS, TypeScript, and ASP.NET Core, including a project management tool deployed to **270** employees and a Welder Management system
- Developed desktop applications DesignCalcs and ProWrite for pressure vessel design, adding features, fixing bugs, and creating unit/regression tests
- Managed CI/CD pipelines, configured testing servers, and ensured smooth system operations
- Operated in an **Agile** development process in a development team of **7** people
- Mentored new recruits, conducted code/design reviews
- Collaborated with clients and team members via **JIRA**

Technologies

Languages: Python, R, C#, Delphi, Java, C/C++, JavaScript, TypeScript, Bash

Technologies: PyTorch, TensorFlow, CARLA Simulator, Docker, Github Actions, ASP.NET Core, React JS, Vue JS, TestComplete, DUnit

Awards

- Won the NC State College of Engineering Graduate Merit Award - *June 2025*
- Won the 1st prize in the inaugural NCStateCS Graduate Research Symposium - *April 2026*